

JAMES HAN

Agronomic Systems Modeling

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SUMMARY

Agricultural scientist with a Ph.D. in crop modeling who implements DSSAT/CROPGRO-Strawberry processes in Python and develops crop models for practical agricultural decisions. Developed and maintained 30 commercial crop models, including corn and soybean, and built CornSoyWater irrigation decision support. Combines Python and R, crop physiology and phenology, field-data validation, management sensitivity analysis and agricultural data engineering to turn model behavior into usable tools and clear technical reports.

SELECTED TECHNICAL PROJECTS

- DSSAT/CROPGRO-Strawberry in Python: Implemented DSSAT/CROPGRO-Strawberry processes in Python to simulate thermal-time phenology, canopy photosynthesis, biomass allocation and fruit development; built utilities to read DSSAT inputs and compare Python outputs with the Fortran model.
- Blueberry maturity detection: Built a crop-imagery application with YOLO, CPU inference in ONNX Runtime, batch processing, maturity counts and CSV export; connected model development to a usable web tool.

TECHNICAL SKILLS

Crop modeling: DSSAT/CROPGRO-Strawberry (Python implementation), APSIM (workshop and hands-on experience), process-based simulation, corn and soybean phenology, soil-water balance, nutrient and residue processes, management sensitivity, calibration and validation, SALUS

Analysis and data: Python, R, Scala, SQL, Spark, mixed models, cross-validation, agricultural APIs, AWS EC2, Azure, QGIS, satellite imagery

Engineering and communication: Git, GitHub pull requests, Confluence, AI-assisted development with Claude Code, analytical slide decks and written model reports

SELECTED INDUSTRY EXPERIENCE

Quantitative Agricultural Scientist

CIBO Technologies | Aug 2021 - Present

- Evaluated SALUS sensitivity to tillage depth and management, soil temperature and moisture, and planting and harvest dates to explain crop maturity and simulated soil responses.
- Integrated manure parameters into SALUS, explored nitrogen-related outputs and tested residue parameters; compared simulated and measured residue-related outputs to assess model behavior and identify limitations.
- Built Python and Scala workflows on AWS to clean, quality-check and tabulate agricultural data for Verra verification; prepared analysis and reporting inputs from farm-management records and field boundaries.
- Created scripts to ingest trial soil-moisture, temperature and CO2 observations; documented analyses in Confluence and contributed code through GitHub pull requests.

INDUSTRY EXPERIENCE CONTINUED

FieldNET Advisor Solutions Specialist (Data Scientist)

Lindsay Corporation | Aug 2018 - Jul 2021

- Developed and maintained 30 crop models, including corn, soybean and cotton, for the commercial FieldNET Advisor irrigation platform; prototyped in Python and used R cross-validation to evaluate performance. Modified and improved FieldNET Advisor software, including API integration with John Deere Operations Center.
- Combined crop simulations and satellite imagery to correct phenology estimates; built an Azure CNN workflow to identify emergence and flowering from multispectral imagery.
- Led R&D field trials measuring soil moisture, phenology and yield, supervised interns, and translated model findings into reports for product and agronomy teams; co-invented a patented irrigation-scheduling method.

EARLIER INDUSTRY EXPERIENCE

Data Scientist and Software Engineer

Novedea / Colaberry / QCI consulting | Feb 2017 - Aug 2018

Client projects for Climate Corporation, DuPont Pioneer and Monsanto.

- Improved crop growth-stage prediction error from approximately +/-7 days to +/-3 days for a model deployed in a Climate Corporation product; analyzed soybean phenology and environmental drivers.
- Evaluated genotype-by-environment effects in multi-location trials using mixed models and variance components; automated feature selection and quality checks to support repeatable agronomic analysis.
- Processed gigabyte-scale field, hybrid-trial and climate datasets from agricultural APIs and data warehouses using Spark, Python and R on AWS EC2; created spatial analyses with QGIS and Tableau.
- Delivered weekly slide briefings to leadership, explaining modeling results, agronomic implications and project recommendations.

DOCTORAL RESEARCH

Graduate Research Assistant in Crop Modeling

University of Nebraska-Lincoln | 2013 - 2016

- Developed CornSoyWater, a web application for corn and soybean irrigation decisions, connecting process-based crop simulation with weather, soil and field-management inputs.
- Evaluated model performance using five years of field data and a two-year irrigation experiment; examined soil-water balance, crop growth, root distribution, water uptake and yield under deficit irrigation.

SELECTED PUBLICATION AND PATENT

Han, C., Yang, H., Payyala, D., et al. (2025). CornSoyWater, a web application for irrigation decision support on corn and soybean. *Agricultural Water Management*, 313, 109454.
[doi:10.1016/j.agwat.2025.109454](https://doi.org/10.1016/j.agwat.2025.109454)

Charling, K. A., and Han, C. J. (2023). System and method for adjusting irrigation system scheduling based on estimated soil water depletion. U.S. Patent 11,771,025. Lindsay Corporation.

EDUCATION

Ph.D., Agronomy and Horticulture (Crop Modeling) | 2016

University of Nebraska-Lincoln

M.S., Agronomy (Weed Science) | 2012

University of Nebraska-Lincoln